

Programme Information

**Bachelor of Computer Engineering
with Honours**

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1.0 Introduction

Computer Engineering is an engineering discipline that is growing rapidly and in importance in the age of computing platform. Computer platforms are terms used to define the eras of computer and its applications. The computer era started in the 1950's with mainframe computers which were then replaced with servers in the 1980's. The current computing platform that we live in had started in the 2010's is defined as an interaction between mobile computing, social media, cloud computing, big data analytics and Internet-of-Things. In the current computing era, Computer Engineering contribution is not limited to areas with applications such as computer systems and telecommunication system, but it can also be extended into areas of applications such as in mobile computing, manufacturing, medicine, information technology and many more. Unlike other engineering disciplines, Computer Engineering is very dynamic and becoming more pervasive in the world.

Computer engineers are involved in the design, building, testing and development of high technology devices ranging from the most powerful supercomputers to the smallest, most energy efficient microprocessors and microcontrollers used in systems with applications in areas such as social media search engines, data farms, cloud computing systems, virtual reality systems, massively parallel online systems which are used in gaming systems and large data processing systems. In addition to the previously mentioned application areas, Computer engineers also contribute in the automation for electromechanical systems and electronic control systems applied in process plants, automotive industry, aerospace, and even maintenance through new technologies such as Internet-of-Things and robotics. In other words, Computer engineers are in high demand in various fields in the current and future workplace.

The Bachelor of Computer Engineering with Honours programme at Universiti Teknologi PETRONAS emphasises on a strong foundation in physics, mathematics, and programming skills, followed by a thorough coverage of basic electrical and electronic engineering courses such as electrical technology, analogue electronics, digital electronics, microprocessor, and computer architectures. At higher levels, students are exposed to data and computer networking, operating systems, embedded systems, big data analytics and parallel and distributed computing. In the final year, students have the opportunity to major in one of these selected areas of their interest:

- **Intelligent Vision System**
- **Next Generation Communication System**
- **Intelligent Robotics**

1.1 Programme Educational Objectives

- i. Engineers who are competent with the potential to become leaders in Electronics, Computing, ICT and related industries.
- ii. Engineers who are involved towards sustainable development in Electronics, ICT and related industries for betterment on the society.

1.2 Programme Outcomes

To produce graduates with the following outcomes:

- i. **Engineering Knowledge** – Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialisation as specified in WK1 to WK4 respectively to the solution of complex engineering problems.
- ii. **Problem Analysis** – Identify, formulate, conduct research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences (WK1 to WK4).
- iii. **Design/Development of Solutions** – Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations (WK5).

- iv. **Investigation** – Conduct investigation of complex engineering problems using research-based knowledge (WK8) and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
- v. **Modern Tool Usage** – Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations (WK6).
- vi. **The Engineer and Society** - Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems (WK7).
- vii. **Environment and Sustainability** – Understand and evaluate the sustainability and impact of professional engineering work in the solutions of complex engineering problems in societal and environmental contexts (WK7).
- viii. **Ethics** – Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice (WK7).
- ix. **Individual and Teamwork** – Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
- x. **Communication** - Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instruction
- xi. **Project Management and Finance** – Demonstrate knowledge and understanding of engineering management principles and economic decision making and apply these to one's own work, as a member and leader in a team, to manage projects in multidisciplinary environments.
- xii. **Lifelong Learning** - Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

1.3 Graduation Requirements

In order to graduate with the Bachelor of Computer Engineering with Honours, students are required to obtain a minimum of **144 credit hours** and a minimum CGPA of 2.00

Summary of Courses Required for Graduation	
Course Group (G)	Number of Credit Hours
NR - National Requirement	10
UR - University Requirement	15
CC- Core Common	21
CD - Core Discipline	75
CI - Core Industrial Internship	14
CSp - Core Specialisation	9
TOTAL	144

2.0 Programme Curriculum Structure: Bachelor of Computer Engineering with Honours

SEMESTER 1				SEMESTER 2				SEMESTER 3			
Code	Courses	Cr	G	Code	Courses	Cr	G	Code	Courses	Cr	G
MPU1	Penghayatan Etika dan Peradaban (Local) BM Komunikasi 2 (International)	2	NR	MPU1	Falsafah dan Isu Semasa (Local) Malaysian Studies 3 (International)	2	NR	MPU2	One (1) U2 Course	2	NR
PFB1012	Introduction to Oil & Gas Industry & Sustainable Development	2	UR	CFB1032	Health, Safety & Environment	2	CC	LFB1042	Academic Writing	2	UR
HFB2152	Ethics and Integrity	2	UR	FFM1023	Engineering Mathematics 2	3	CC	HFB1023	Scientific Inquiry	2	UR
FFM1013	Engineering Mathematics 1	3	CC	TFB1033	Object Oriented Programming	3	CD	TFB1103	Software Engineering	3	CD
EFB1023	Structured Programming and Interfacing	3	CD	EFB1054	Digital Electronics	4	CD	TFB1113	Discrete Mathematics	3	CD
EFB1013	Circuit Theory	3	CD					KXXXXX1	Co Curriculum I	1	UR
Credit Hours		15		Credit Hours		14		Credit Hours		13	
SEMESTER 4				SEMESTER 5				SEMESTER 6			
Code	Courses	Cr	G	Code	Courses	Cr	G	Code	Courses	Cr	G
MPU 3	One (1) U3 Course	2	NR	FFM2063	Data Analytics	3	CC	MPU 4	Community Engagement Project	2	NR
HFB2033	Professional Comm. Skills	3	UR	MFB2061	Engineering Team Project	1	CC	KXXXXX1	Co Curriculum II	1	UR
EFB1043	Signals and Systems	3	CD	RBB2042	Linear Algebra and Matrix Methods	2	CD	RBB3013	Instrumentation and Control	3	CD
TFB2023	Algorithm and Data Structure	3	CD	GFB2102	Entrepreneurship	2	UR	MFB2062	Engineering Team Project	2	CC
TFB2013	Operating Systems	3	CD	EFB2043	Communication Systems	3	CD	EFB2073	Microprocessor and Computer Architecture	3	CD
								RBB2013	Digital Twin	3	CD
Credit Hours		14		Credit Hours		11		Credit Hours		14	

SEMESTER 7				STUDENT INDUSTRIAL INTERNSHIP PROGRAMME				STUDENT INDUSTRIAL INTERNSHIP PROGRAMME			
Code	Courses	Cr	G	Code	Courses	Cr	G	Code	Courses	Cr	G
RBB3023	Embedded and IOT Systems	3	CD	IFB3037	Student Industrial Training	7	CI	IFB3047	Student Industrial Project	7	CI
EFB4133	Digital System Design	3	CD								
EFB3043	Data & Computer Network	3	CD								
RBB3033	Cloud System Technologies	3	CD								
Credit Hours		12		Credit Hours		7		Credit Hours		7	
SEMESTER 8				SEMESTER 9				SEMESTER 10			
Code	Courses	Cr	G	Code	Courses	Cr	G	Code	Courses	Cr	G
VEB3102	Project Management	2	CC	GFB3022	Engineering Economics	2	CC	EFB4073	Engineers in Society	3	CC
EFB4033	Probability and Random Process	3	CD	EFB4063	Artificial Intelligence & Applications	3	CD	RBB4044	Final Year Project II	4	CD
RBB4013	System Integrated Design Project	3	CD	RBB4032	Final Year Project I	2	CD	RBB4603	Distributed and Parallel Processing	3	CD
RBB4XX3	Core Specialisation	3	CSp	RBB4XX3	Core Specialisation	3	CSp	RBB4XX3	Core Specialisation	3	CSp
TFB2113	Computer Security	3	C								
Credit Hours		14		Credit Hours		10		Credit Hours		13	
Total Credit Hours : 144											

3.0 Core Specialisation (CSp) Courses

Choose any of the following sets of Core Specialisations areas (set A or B) for Core Specialisation I, II and III.

A. Intelligent Vision System				
No	Code	Course Name	Cr	G
1	RBB4613	Intelligent Systems	3	CSp
2	RBB4623	Image Processing and Machine Vision	3	CSp
3	RBB4633	Next Generation Cybersecurity	3	CSp
B. Next Generation Communication System				
No	Code	Course Name	Cr	G
1	RBB4713	Next Generation Communications and Networks	3	CSp
2	RBB4723	Wireless Sensor Networks	3	CSp
3	RBB4733	Digital Communications	3	CSp
C. Intelligent Robotics				
No	Code	Course Name	Cr	G
1	RBB4813	Robotics and Automation	3	CSp
2	RBB4623	Image Processing and Machine Vision	3	CSp
3	RBB4823	Autonomous Robot Systems	3	CSp

4.0 Minor (EM) Courses

Minor in Management is made available to all Engineering students who are interested to enhance their knowledge in management and business. However, the option to do the Minor in Management is at the students' discretion and is **not compulsory**. The credit hours will be over and above the graduation requirements of **145 credit hours**.

Minor in Management					
No	Code	Courses	Cr	G	Semester
1	GDB2013	Business Accounting	3	EM	Students can take these courses in any semester
2	GDB1033	Management and Organizational Behaviour	3	EM	
3	GDB2053	Principles of Finance	3	EM	
4	GDB2043	Principles of Marketing	3	EM	
5	GDB3013	Small Business and Entrepreneurship	3	EM	
Total Credit Hours			15		

Students have to complete and pass all of listed courses to get Minor in Management. All results will be counted in the GPA/CGPA calculation.

Note:

Code A	:	Please refer to Materials Engineering Programme for Course Synopsis.
Code C	:	Please refer to Chemical Engineering Programme for Course Synopsis.
Code E	:	Please refer to Electrical & Electronic Engineering and Computer Engineering Programme for Course Synopsis.
Code F, Y, Z	:	Please refer to Applied Chemistry, Applied Physics Programme and Fundamental & Applied Sciences Department for Course Synopsis
Code K	:	Please refer to Credited Co-Curriculum Programme for Course Synopsis.
Code M	:	Please refer to Mechanical Engineering Programme for Course Synopsis.
Code P	:	Please refer to Petroleum Engineering Programme for Course Synopsis.
Code Q	:	Please refer to Petroleum Geoscience Programme for Course Synopsis.
Code T	:	Please refer to Computer Science, Information Systems and Information Technology Programmes for Course Synopsis.
Code V	:	Please refer to Civil Engineering Programme for Course Synopsis.
Code L, G, H, MPU	:	Please refer to Business Management Programme and Management & Humanities Department for Course Synopsis.

5.0 Programme Courses Synopsis

5.1 Core Discipline (CD) Courses Synopsis

COURSE CODE : EFB1023
COURSE : Structured Programming and Interfacing
NO. OF CREDIT : 3 credits
PRE-REQUISITE : None

This programming course is designed to allow electrical and electronics students to design and develop computer programs to solve electronics systems problems. The course will adopt a lecture/hands-on approach. Using this approach, students will explore a structured programming language. Students will be required to design and develop computer programs to control external devices via the PC.

COURSE CODE : EFB1013
COURSE : Circuit Theory
NO. OF CREDIT : 3 credits
PRE-REQUISITE : None

This course introduces the fundamental concepts related to the principles of electrical engineering, the quantities, units, symbols, definitions in the field of electrical engineering, the passive components RLC, the techniques of analysing electrical circuits, the concepts of power, the two and three phases' circuits. In addition, the course will also cover fundamental operation of operational amplifiers and their application.

COURSE CODE : TFB1033
COURSE : Object Oriented Programming
NO. OF CREDIT : 3 credits
PRE-REQUISITE : Structured Programming and Interfacing

This course introduces fundamental concepts of object-oriented programming such as encapsulation, inheritance, and polymorphism. Students will apply the concepts to build software system that are highly reusable and extensible.

COURSE CODE : EFB1054
COURSE : Digital Electronics
NO. OF CREDIT : 4
PRE-REQUISITE : None

This course covers topic related to the principles and techniques of designing of digital logic circuit. It introduces the binary number system, Boolean algebra, digital circuits for performing digital arithmetic operations. On more advanced topics, it covers the design of sequential circuits which includes state assignment, state minimization, and detection and elimination of hazards that exist in digital circuits.

COURSE CODE : TFB1103
COURSE : Software Engineering
NO. OF CREDIT : 3
PRE-REQUISITE : None

Software Engineering addresses the theory and methodology of programming of enormous and heterogeneous computer software. The course covers project planning and management, requirements identification, analysis, and management, development tools and methods, UML, software architecture and design, detailed design, implementation, and testing. Special emphasis is given to planning and estimating and to quality development practices such as product inspections and testing and an in-depth understanding of why computers are essential components in the business world and society in general. The courses will also focus on the computer as a valuable productivity tool and recognition of the personal computer's position as the backbone of the computer industry. The course would also highlight on the importance of computers utility as a standalone and/or within networked devices

COURSE CODE : TFB1113
COURSE : Discrete Mathematics
NO. OF CREDIT : 3
PRE-REQUISITE : None

This course introduces the fundamental concepts of Discrete Mathematics such as sets, functions, counting principles, logic, proof techniques, mathematical induction, graphs, directed graphs, trees, relations and Boolean algebra. Students will be exposed to problem-solving skills by applying the theoretical concepts of Discrete Mathematics.

COURSE CODE : EFB1043
COURSE : Signals and Systems
NO. OF CREDIT : 3
PRE-REQUISITE : Circuit Theory

This course covers topics related to theoretical mathematical foundation of signals and systems. The focuses are on mathematics and practical issues of signals in continuous and discrete time, linear time-invariant systems, convolution, Fourier transforms and Laplace transforms. MATLAB is used as a learning tool.

COURSE CODE : TFB2023
COURSE : Algorithm and Data Structure
NO. OF CREDIT : 3
PRE-REQUISITE : None

This course examines various data structures for storing and accessing data, and algorithms for efficiently finding solutions to problems involving the data. It emphasizes the ability to solve a programming problem by first designing appropriate data structures and deriving provably effective and efficient algorithms. Students learn to theoretically analyse algorithms and to empirically measure their implementations.

COURSE CODE : TFB2013
COURSE : Operating Systems
NO. OF CREDIT : 3
PRE-REQUISITE : None

This module is intended to provide students with a sound understanding of the concepts relating to the design and implementation of the computer operating systems.

COURSE CODE : RBB2042
COURSE : Linear Algebra and Matrix Methods
NO. OF CREDIT : 2
PRE-REQUISITE : None

This course covers System of Linear Equations, Vector Spaces, Matrix, Determinants, Matrix Bases, Eigenvalues and Eigenvectors, System of Differential Equations, Orthogonality Projections, Matrix Decomposition and programming using software tools.

COURSE CODE : EFB2043
COURSE : Communication Systems
NO. OF CREDIT : 3
PRE-REQUISITE : Signal and Systems

This course serves as an introduction to familiarize students with fast expanding market of communication systems. Emphasis is given to digital modulation schemes comprising a multiplexer/transmitter, a communication channel and its associated noise, interference and distortion effects, and a demultiplexer/receiver.

COURSE CODE : RBB3013
COURSE : Instrumentation and Control
NO. OF CREDIT : 3
PRE-REQUISITE : Circuit Theory

This course is designed to equip computer engineers with the fundamental of instrumentation and measurement systems and introduces the concept of digital control system. Knowledge and familiarity in sensors, signal conditioning and processing and the associated feedback control for which the implementation is performed using digital computers are the important elements of this course.

COURSE CODE : EFB2073
COURSE : Microprocessor and Computer Architecture
NO. OF CREDIT : 3
PRE-REQUISITE : Digital Electronics

This course covers topics related to microprocessor technology which include the principle architecture of a microprocessor/microcontroller, the instruction set architecture and machine codes, the system hardware design, and the programming tasks.

COURSE CODE : RBB2013
COURSE : Digital Twin
NO. OF CREDIT : 3
PRE-REQUISITE : None

This course will introduce two key areas which is matrix computing and physics-based numerical models which are important to develop digital twin models. Students will learn about random variables, apply concepts of linear algebra which covers vector spaces and matrix methods in computing solutions to finite element and finite difference models with practical examples applied to physics-based systems. Students will also learn how to use R or Python for numerical modelling of digital twin models.

COURSE CODE : RBB3023
COURSE : Embedded and IOT Systems
NO. OF CREDIT : 3
PRE-REQUISITE : Microprocessor and Computer Architecture

This course explores practical application of microcontrollers and design of embedded systems for advanced applications. It also covers concept real-time operating systems and their practical applications.

COURSE CODE : EFB4133
COURSE : Digital System Design
NO. OF CREDIT : 3
PRE-REQUISITE : Microprocessor and Computer Architecture

This course covers design, validation, and testing techniques for digital. Students will learn how to design both individual building blocks as well as assemble digital systems from combinations of these blocks and analyse digital systems.

COURSE CODE : EFB3043
COURSE : Data and Computer Network
NO. OF CREDIT : 3
PRE-REQUISITE : Communication Systems

This course introduces and explains the protocols, services and standards used in multimedia networks, which is actually the integration of voice/data communications, and computer networks. It covers topics on data communications principles and techniques such as data communications model, physical layer interfaces, and data link protocols for error detection and correction. It also covers topics on networking such as concept of layered TCP/IP and OSI protocols, local area networks (Ethernet, token ring, etc) and wide area networks

COURSE CODE : RBB3033
COURSE : Cloud System Technologies
NO. OF CREDIT : 3
PRE-REQUISITE : None

This is a course that exposes students to the setup, configuration and administration of the fundamental networking system that enable students to create a network and subnetwork for large data storage and data access. In addition to development of networks, students will also be introduced to scaling up, optimizing, and measuring the operations of cloud storage systems. This will be supplemented with tools and methodologies for visually extracting and analysing the performance of these cloud systems.

COURSE CODE : EFB4033
COURSE : Probability and Random Processes
NO. OF CREDIT : 3
PRE-REQUISITE : None

This course introduces the students to the concept and statistical properties of random variables – both discrete and continuous - for several types of distributions. Students will then analyze the significance of correlation and covariance between two random variables. The course also considers properties and problem solutions of stationary and random processes.

COURSE CODE : RBB4013
COURSE : System Integrated Design Project
NO. OF CREDIT : 3
PRE-REQUISITE : None

The System Integrated Design Project (SIDP) integrates current and previous knowledge and skills in engineering, project management, and business disciplines. The course emphasizes practical, hands-on experience, and integrates analytical and design skills. Students will develop skills in problem solving, creative thinking, project planning, and teamwork. Significant experience in developing, designing, prototyping, proving, and verifying their design will also be attained by the students. Engineering communication, such as reports, and oral presentations are also covered.

In SIDP, students work in teams under the direction and guidance of a faculty advisor to tackle a challengingly complex prototype-based project.

COURSE CODE : EFB4063
COURSE : Artificial Intelligence and Applications
NO. OF CREDIT : 3
PRE-REQUISITE : None

The students will be introduced with the basic concepts of the topics. The students will be guided to program the algorithms and apply on case studies.

COURSE CODE : RBB4032
COURSE : Final Year Project I
NO. OF CREDIT : 2
PRE-REQUISITE : None

In the final year project, students will apply and put into practice all the skills and technical knowledge gained throughout the Electrical and Electronic Engineering program to design and implement a solution to a research or practical problem.

COURSE CODE : RBB4044
COURSE : Final Year Project II
NO. OF CREDIT : 4
PRE-REQUISITE : Final Year Project I

This is an individual research project in connection with a special engineering problem and under the guidance of a faculty member. The project undertaken may fall under one of the following areas: mathematical analysis, experimental tests, computer simulation, hardware and/or software development, device fabrication. For both FYP I and II, each student prepares a comprehensive engineering report, present and demonstrate findings and results of the project work.

COURSE CODE : RBB4603
COURSE : Distributed and Parallel Processing
NO. OF CREDIT : 3
PRE-REQUISITE : Microprocessor and Computer Architecture

In this course, students will learn how to analyse a programming task and design parallel programs to execute this task. The course will cover the different models of data access in parallel and distributed computing systems. Students will study the common computation patterns in large datasets, strategies for parallelizing these computations and software design patterns commonly used to engineer parallel programs. Students will also learn how to use measurement and debugging tools for parallel programs.

5.2 Core Specialisation (CSp) Courses Synopsis

5.2.1 Intelligent Vision System

COURSE CODE : RBB4613
COURSE : Intelligent Systems
NO. OF CREDIT : 3
PRE-REQUISITE : None

The course provides a coverage of the principles and techniques in deep neural network artificial intelligence. Students will learn the principles of deep neural network artificial intelligence and build deep neural network applications. Students will also learn to describe model performance and datasets in a complete manner and be able to assess energy and memory resource requirements. Finally, students will consider security and privacy issues surrounding deep neural networks.

COURSE CODE : RBB4623
COURSE : Image Processing and Machine Vision
NO. OF CREDIT : 3
PRE-REQUISITE : None

The course deals with the theory, principles and techniques of image processing and computer vision, which includes pre-processing images to improve image quality, image filtering and restoration, image feature descriptors, image analysis and transforms. The practical aspects of the course would cover the problem solving and application of algorithms to practical problems.

COURSE CODE : RBB4633
COURSE : Next Generation Cybersecurity
NO. OF CREDIT : 3
PRE-REQUISITE : None

This course covers advance topics related to computing security and current trends in related with computing security. This course is an extension of computing security which provide fundamental knowledge on network security. The course will cover advanced topics in computer networks such as vulnerabilities in different types of networks, detection methods, and “state-of-the-art” techniques to prevent them. The course would also cover computer traffic analysis and network devices/tools used in security technology. Finally, the course would also cover next generation topics in the field of cybersecurity, including advanced cryptography, networking, cyber defence etc.

5.2.2 Next Generation Communication System

COURSE CODE : RBB4713
COURSE : Next Generation Communications and Networks
NO. OF CREDIT : 3

PRE-REQUISITE : None

This course provides an introductory concept of Wireless Communication and Networks. While the focus is to give a broad overview of emerging topics of Wireless communications and networks, it also provides some theme examples that enhance students' knowledge in the areas of the Next Generation Communication Systems.

COURSE CODE : RBB4723
COURSE : Wireless Sensor Networks
NO. OF CREDIT : 3
PRE-REQUISITE : None

Recent advances in wireless communications and electronics have enabled the development of low-cost, low-power, multifunctional sensor nodes that are small and these nodes communicate untethered in short distances. These tiny sensor nodes, which consist of sensing, data processing, and communicating components, leverage the idea of sensor networks. The sensor networks can be used for various application areas, e.g., health, military, home, etc.

COURSE CODE : RBB4733
COURSE : Digital Communications
NO. OF CREDIT : 3
PRE-REQUISITE : None

This course is a formal introduction to the main concepts of digital communications. Topics covered include: baseband pulse transmission, passband digital transmission, signal-space representation and information theory such as channel coding, source coding and capacity. Applications are drawn from the access network parts of the telephone networks.

5.2.3 Intelligent Robotics

COURSE CODE : RBB4813
COURSE : Robotics and Automation
NO. OF CREDIT : 3
PRE-REQUISITE : None

This course will introduce the world and knowledge of robotics which covers the areas of robot dynamics, PLC, sensors and actuators that will be used for automation of systems.

COURSE CODE : RBB4823
COURSE : Autonomous Robot Systems
NO. OF CREDIT : 3
PRE-REQUISITE : None

This course will introduce the utilization and implementation of autonomous movement and navigation of robots which could be autonomous ground or aerial robots.

COURSE CODE : RBB4623
COURSE : Image Processing and Machine Vision
NO. OF CREDIT : 3
PRE-REQUISITE : None

The course deals with the theory, principles and techniques of image processing and computer vision, which includes pre-processing images to improve image quality, image filtering and restoration, image feature descriptors, image analysis and transforms. The practical aspects of the course would cover the problem solving and application of algorithms to practical problems.